

DS Practical - 8

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Write programs to implement and compare the following sorting algorithms:

a. Bubble sort

CODE :

```
def bubbleSort(array):
    for i in range(len(array)):
        for j in range(0, len(array) - i - 1):
            if array[j] > array[j + 1]:

                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp

data = [-2, 45, 0, 11, -9]
bubbleSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("KOMAL S049")
```

OUTPUT :

```
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
KOMAL S049
```

b. Insertion sort

CODE :

```

def insertionSort(array):

    for step in range(1, len(array)):
        key = array[step]
        j = step - 1
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        array[j + 1] = key

data = [9, 5, 1, 4, 3]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("KOMAL S049")

```

OUTPUT :

```

Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]
KOMAL S049

```

c. Selection sort

CODE :

```

def bubbleSort(array):
    for i in range(len(array)):
        for j in range(0, len(array) - i - 1):
            if array[j] > array[j + 1]:

                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp

data = [-2, 45, 0, 11, -9]
bubbleSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("KOMAL S049")

```

OUTPUT :

```

Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]
KOMAL S049

```